(17)

a) $H_0: \sigma_1^2 = \sigma_2^2$

$H_a: \sigma_1^2 \neq \sigma_2^2$

$$F^* = \frac{S_1^2}{S_2^2}$$

$$= \frac{2.78095}{0.17143}$$

$$= 16.22$$

$$t_{df_1, df_2, \frac{\alpha}{2}} = t_{14,14,0.025}$$

$$= 2.9786$$

$$F^* = 16.22 > t_{14,14,0.025} = 2.9786$$

$\therefore$ reject H_0

b) $H_0: \mu_1 - \mu_2 = 0$

$H_a: \mu_1 - \mu_2 \neq 0$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

$$= \frac{3.73 - 4.80}{\sqrt{\frac{2.78095}{15} + \frac{0.17143}{15}}}$$

$$= \frac{-1.07}{0.4417}$$

$$= -2.4211$$

$$df = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{(S_1^2/n_1)}{(n_1-1)} + \frac{(S_2^2/n_2)}{(n_2-1)}}$$

$$= \frac{(0.1854 + 0.0114)^4}{\frac{0.1854^2}{14} + \frac{0.0114^2}{14}}$$

$$= \frac{0.03873}{0.002464}$$

$$= 15.72$$

$$\approx 16$$

$$|t| = t_{df, \frac{\alpha}{2}} = t_{16, 0.025}$$

$$\approx 2.12$$

$\because |-2.4211| = 2.41 > 2.12$

$\therefore$ reject H_0

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a) $H_0: \mu_2 - \mu_1 = 0$

$H_a: \mu_2 - \mu_1 < 0$

$$S_p^2 = \frac{(n_1-1)S_1 + (n_2-1)S_2}{n_1+n_2-2}$$

$$= \frac{4(0.025) + 4(0.013)}{5+5-2}$$

$$= 0.019$$

$$S_p = \sqrt{0.019}$$

$$= 0.1378$$

$$t^* = \frac{\bar{x}_2 - \bar{x}_1}{S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}}$$

$$= \frac{-0.14}{0.08718}$$

$$= -1.606$$

$df = 8$

$t_{8,0.05} = -1.86$

$\therefore t^* = -1.606 > t_{8,0.05} = -1.86$, non-reject H_0

$$\bar{x}_1 = \frac{4.1+5.2+5+4.9+5.1}{5}$$

$$= 5$$

$$S_1^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})}{n-1}$$

$$= \frac{0.1}{4}$$

$$= 0.025$$

$$\bar{x}_2 = \frac{4.7+4.9+4.8+4.9}{4}$$

$$= 4.86$$

$$S_2^2 = \frac{0.052}{4}$$

$$= 0.013$$

b) $(\bar{x}_1 - \bar{x}_2) \pm t_{8,0.025} S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}$

$$= 5 - 4.86 \pm 2.306(0.08718)$$

$$= 0.14 \pm 0.201$$

$$= (-0.061, 0.341)$$

a) $H_0: \mu_d = 0$

$H_a: \mu_d \neq 0$

$$\bar{d} = \frac{0.1+0.1+0+\cancel{(0.2+0.1)}\ 0.2-0.1}{5}$$

$$= 0.06$$

$$S_d^2 = \frac{\left(\sum_{i=1}^{n}(d_i - \bar{d})^2\right)}{n-1}$$

$$S_d = \sqrt{0.013}$$

$$= 0.0114$$

$$SE = \frac{S_d}{\sqrt{n}}$$

$$= \frac{0.114}{\sqrt{5}}$$

$$= 0.051$$

$$t^* = \frac{\bar{d} - D}{SE}$$

$$= \frac{0.06}{0.051}$$

$$= 1.18$$

$df = 4$

$\therefore t_{4,0.025} = 2.776 > t^* = 1.18$, non-reject H_0

$$\bar{d} \pm t_{4,0.025} \cdot SE = 0.06 \pm 2.776(0.051)$$

$$= (-0.076, 0.216)$$

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(20)

$H_0: \mu_1 = 0$

$H_a: \mu_1 > 0$

$n = 20$

$$\bar{d} = \frac{15 \times 3 + 9 + 16 + 8 + 12 + 8 + 10 + 12 + 12 + 11 + 4 + 10 + 4 + 17 + 13 + 4 + 7 \times 2 + 10}{20}$$

$= 10.25$

$S_d = \sqrt{\frac{\sum_{i=1}^{n}(d_i - \bar{d})}{n-1}}$ $\quad SE = \frac{S_d}{\sqrt{n}}$ $\quad t^* = \frac{\bar{d} - 0}{SE}$ $\quad df = 19$

$= 4.051$ $\quad = 0.906$ $\quad = 11.31$

$\therefore t_{19, 0.05} = 1.729 < t^* = 11.31$

~~non reject H_0~~ reject H_0

$\bar{d} \pm t_{19, 0.025} SE = 10.25 \pm \cancel{1.729}\ 2.093 (0.906)$

$= 10.25 \pm 1.946$

$= (8.304, 12.196)$

22) a) $\bar{d} = \frac{-0.006 - 0.018 - 0.014 - 0.011 - 0.004 - 0.014}{6}$

$= -0.01$ $\quad SE = \frac{S_d}{\sqrt{n}}$ $\quad \therefore t_{5, 0.025} = 2.571 > |t^*| = 2.57$

$S_d = \sqrt{\frac{\sum_{i=1}^{n}(d_i - \bar{d})^2}{n}}$ $\quad = 0.00784$ $\quad \therefore$ non-reject H_0.

$= 0.00953$ $\quad t^* = \frac{\bar{d} - 0}{SE}$

$= \frac{-0.01}{0.0384}$

$= -2.57$

$H_0: \mu_d = 0$

$H_a: \mu_d \neq 0$

$df = 5$

b) $\bar{d} \pm t_{5, 0.025} SE$

$= -0.01 \pm 2.571 (0.00784)$

$= (-0.02, 0)$

(3)